

Each contribution answers one question with its own protocol

| Contribution                                | Data                                             | Protocol                                         | Metrics                                                 | Reported in |
|---------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------------|-------------|
| <div>C1</div> <div>Representation</div>     | KDDTrain <sup>+</sup><br>UNSW train              | Three-stage rule:<br>$V \geq T$ , KTA, min $Q$   | $V(n)$ , KTA( $n$ ),<br>$Q(n)$ , $n^*$                  | Figs. 4-5   |
| <div>C2</div> <div>Entanglement</div>       | $N_{\text{train}} = 1000$<br>10 runs             | ZZ vs Z ablation,<br>paired within run           | $\Delta\text{KTA}$ , $\Delta F_1$ ,<br>noise validation | Figs. 6-7   |
| <div>C3</div> <div>Distribution shift</div> | KDDTest <sup>+</sup> ,<br>KDDTest <sup>-21</sup> | Prior shift, perturbation,<br>attack composition | $\Delta F_1$ , Holm,<br>McNemar                         | Fig. 8      |
| <div>C4</div> <div>Sample complexity</div>  | Nested subsets<br>$N = 10^2\text{-}10^4$         | Two sampling regimes,<br>two tuning arms         | Learning curve,<br>crossover, rare $F_1$                | Figs. 9, 11 |